

Project Report

FinWise AI

Intelligent Loan Eligibility, Credit Analysis & Financial Advisory Platform

Domain: Finance (BFSI)

Submitted by

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1. Introduction

1.1 Project Overview

FinWise AI is an AI-powered BFSI (Banking, Financial Services & Insurance) web platform designed to simplify and accelerate personal financial decision-making. It brings together four core financial tools — a Loan Eligibility Checker, a Credit Score Analyzer, an EMI Calculator, and an AI Financial Tips advisor — within a single, seamless interface, so users no longer need to consult multiple disconnected tools or wait on manual bank processes to understand their financial standing.

The platform is built using HTML5, CSS3 (with a dark glassmorphism visual style), and JavaScript on the front end, and integrates Claude AI to power real-time, intelligent financial reasoning. Application data is persisted through a Google Sheets integration, giving users and administrators a transparent, always-available record of submissions and outcomes across sessions.

1.2 Purpose

The purpose of FinWise AI is to remove the friction, delay, and opacity that typically surround loan and credit decisions. Traditional loan eligibility checks often require in-person visits, physical paperwork, and multi-day turnaround times, while credit improvement guidance is rarely personalized. FinWise AI addresses this by:

- Giving users an instant, data-driven estimate of loan eligibility and risk classification.
- Helping users understand and improve their credit score through structured, actionable advice.
- Allowing users to simulate EMI outflows before committing to a loan, supporting informed financial planning.
- Offering a conversational AI advisor for on-demand, personalized financial guidance.
- Maintaining transparent, persistent records of applications via Google Sheets synchronization.

2. Ideation Phase

2.1 Problem Statement

PS	I am (User)	I'm trying to	But	Because	Which makes me feel
PS-1	a salaried employee	check my loan eligibility	I can't get a quick answer	banks require in-person visits and manual document review	uncertain and frustrated
PS-2	a first-time borrower	understand my credit score	I don't know what's hurting it	credit reports are technical and not actionable	confused and discouraged
PS-3	a prospective loan applicant	plan my monthly budget around a new loan	I can't predict the exact EMI	interest and tenure calculations aren't transparent upfront	anxious about affordability

2.2 Empathy Map Canvas

SAYS	THINKS
"Will I even qualify for this loan?" / "Why is my credit score low?"	"I hope the bank doesn't reject me after all this paperwork." / "There must be an easier way to check this myself."
DOES	FEELS
Searches online for eligibility calculators; asks friends/family about EMI amounts; delays applying due to uncertainty.	Anxious about rejection, overwhelmed by financial jargon, relieved when given clear, instant answers.

2.3 Brainstorming

The team brainstormed a wide range of ideas to address the problem statements above, then grouped and prioritized them:

- Instant loan eligibility calculator with risk classification (High Priority)
- Credit score analyzer with component-wise breakdown (High Priority)
- EMI calculator with amortization schedule and CSV export (High Priority)
- Conversational AI financial advisor powered by Claude (High Priority)
- Google Sheets sync for transparent, persistent record-keeping (Medium Priority)
- Admin panel for managing API keys and reviewing application logs (Medium Priority)
- Live market ticker (NIFTY 50 / SENSEX) for contextual awareness (Low Priority)

The four highest-priority ideas were carried forward as the platform's core modules: Loan Eligibility Checker, Credit Score Analyzer, EMI Calculator, and AI Financial Tips.

3. Requirement Analysis

3.1 Customer Journey Map

Awareness	Consideration	Decision	Action	Retention
User lands on the FinWise AI platform seeking loan/credit guidance	Explores Loan Checker, Credit Analyzer, and EMI Calculator to compare options	Submits application profile for instant AI evaluation	Receives approval/rejection, risk rating, and DTI ratio; consults AI Advisory if needed	Returns periodically to re-check score improvements and plan future loans

3.2 Solution Requirement

Functional Requirements:

- Accept applicant income, credit score, existing EMIs, requested loan amount, and tenure as input.
- Compute Debt-to-Income (DTI) ratio and classify credit risk as Low / Medium / High.
- Approve or reject loan applications with a maximum eligible credit limit.
- Analyze credit scores (FICO/CIBIL) and break them down by Payment History, Utilization, Length of History, and Credit Mix.

- Calculate EMI, total interest, and total payable amount with a full amortization schedule, exportable as CSV.
- Provide a conversational AI advisory console for financial queries, powered by Claude.
- Synchronize application records to Google Sheets and expose an admin panel for configuration.

Non-Functional Requirements:

- Instant (near real-time) response for eligibility, credit, and EMI calculations.
- Secure handling of API keys and service account credentials.
- Responsive, accessible UI across devices.
- Data transparency and persistence across sessions via Google Sheets.

3.3 Data Flow Diagram

Level 0: User submits application data (income, credit score, EMIs, loan amount, tenure) → FinWise AI Engine validates and processes the input → Risk & DTI evaluation logic classifies the application → Claude AI generates a natural-language assessment and recommendations → Result is displayed to the user and appended to the local cache.

Level 1: On synchronization, cached application records flow from the local store to the Google Sheets API using stored service account credentials, updating the Sheets Sync status and Database Sync Statistics shown in the Admin panel.

3.4 Technology Stack

Layer / Skill	Technology
Markup & Structure	HTML5
Styling	CSS3 with CSS Animations (dark glassmorphism UI)
Client Logic	JavaScript
Backend / Scripting	Python
AI Reasoning	Generative AI (Claude) for eligibility analysis, credit tips, and advisory chat
Integration	REST APIs, JSON for data exchange
Data Persistence	Google Sheets API (via service account credentials)
Version Control	Git

4. Project Design

4.1 Problem Solution Fit

Users struggle to get fast, transparent answers about loan eligibility, credit health, and repayment affordability, and generic financial advice rarely accounts for their specific numbers. FinWise AI fits this problem by combining rule-based financial calculations (DTI ratio, EMI, credit scoring bands) with generative AI reasoning, so every user receives both a precise numeric result and a personalized, explained recommendation in the same interaction.

4.2 Proposed Solution

Problem	Proposed Solution	Novelty / Uniqueness	Business Model
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Slow, opaque loan and credit evaluation	Unified AI-powered BFSI platform with 4 integrated tools	Claude AI reasoning combined with rule-based risk/DTI engine, in one glassmorphism interface	Freemium for individual users; API/white-label licensing for banks & NBFCs
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4.3 Solution Architecture

FinWise AI follows a three-layer architecture:

- Presentation Layer: HTML5/CSS3/JavaScript single-page interface with dedicated sections for Loan Checker, Credit Analyzer, EMI Calculator, AI Advisory, and Admin.
- Intelligence Layer: Claude AI receives applicant parameters via REST calls and returns JSON-structured risk assessments, credit tips, and advisory responses.
- Data Layer: A local application cache captures every submission; on sync, records are written to a Google Sheet via a service-account-authenticated REST call, keeping a durable, auditable history.

5. Project Planning & Scheduling

5.1 Project Planning

The project was executed across four sprints using story points (Fibonacci scale: 1, 2, 3, 5):

Sprint	Functional Requirement (Epic)	Story Points	Priority
Sprint 1	UI shell, glassmorphism theme, and Loan Eligibility Checker (input form, DTI logic, risk classification)	13	High
Sprint 2	Credit Score Analyzer (FICO/CIBIL bands, component breakdown) and EMI Calculator with amortization schedule	16	High
Sprint 3	Claude AI integration for real-time analysis, AI Financial Advisory chat console, quick-query shortcuts	18	High
Sprint 4	Google Sheets sync, Admin panel, CSV export, testing, and deployment to Netlify	12	Medium

Total Story Points = 59 across 4 sprints → Team Velocity ≈ 14.75 story points per sprint.

6. Functional and Performance Testing

6.1 Performance Testing

The following representative test scenarios were used to validate the Loan Eligibility Checker, Credit Score Analyzer, and EMI Calculator:

Scenario	Test Input	Expected / Actual Result
1. Salaried Individual	Monthly income ₹45,000; credit score 730; no existing EMIs	Approved; eligible loan ₹9,00,000; Risk: Low
2. High-Risk Applicant	Monthly income ₹22,000; credit score 620; existing EMI ₹18,000	Rejected; Risk: High; redirected to Credit Score Analyzer

3. EMI Planning	Loan ₹5,00,000 at 10.5% for 60 months	Monthly EMI, total interest, and total payable computed instantly
4. Credit Improvement	Credit score 580	Classified as Poor; actionable recommendations shown; user directed to AI Financial Tips

All four scenarios returned results instantly (sub-second calculation time for DTI/EMI logic), with Claude AI narrative responses returning within a few seconds, well within acceptable BFSI web-application response benchmarks.

7. Results

7.1 Output Screenshots

The deployed application (<https://sensational-selkie-59db0b.netlify.app/>) demonstrates the following screens in production:

- Loan Eligibility Checker — application profile form with instant AI evaluation, showing Approved/Rejected status, credit risk rating, proposed DTI ratio, applicable interest rate, proposed monthly EMI, and max eligible credit limit.
- Credit Score Analyzer — FICO/CIBIL standing gauge with a weighted breakdown (Payment History 35%, Credit Utilization 30%, Length of History 15%, Credit Mix & New Lines 20%) and actionable improvement steps.
- Loan EMI Calculator — interactive sliders for principal, interest rate, and tenure, a payment summary (monthly EMI, total interest, total payable), and a full amortization breakup schedule exportable as CSV.
- AI Financial Advisory Console — a chat interface powered by Claude AI with quick-query shortcuts (credit score improvement, debt consolidation, 50/30/20 budgeting, emergency buffer setup).
- Admin Panel — Google Sheets integration parameters, database sync statistics, and a local applications log table.

[Insert screenshots of each screen here from the live deployment before final submission.]

8. Advantages & Disadvantages

Advantages

- Instant, transparent loan eligibility and credit risk decisions without manual bank processing delays.
- Combines rule-based financial computation with generative AI explanations for more understandable results.
- Single unified platform for eligibility checking, credit analysis, EMI planning, and financial advisory.
- Persistent, auditable record-keeping via Google Sheets synchronization.

- Actionable, structured guidance for users looking to improve a poor credit score.

Disadvantages

- Risk and eligibility logic is currently rule-based on limited parameters and does not yet incorporate a full bureau-grade credit history.
- Dependence on third-party services (Claude AI API, Google Sheets API) introduces external service availability risk.
- Client-side storage of API keys/service account credentials in the Admin panel requires stronger secrets management for production use.
- No user authentication layer yet, so application history is not tied to a verified individual identity.

9. Conclusion

FinWise AI successfully demonstrates how generative AI can be combined with straightforward financial calculations to make loan eligibility checks, credit analysis, EMI planning, and financial advisory instantly accessible in one interface. Across the tested scenarios — from a low-risk salaried applicant to a high-risk rejection case, EMI planning, and credit score improvement — the platform consistently produced accurate, explainable, and actionable results, validating the core problem-solution fit identified during the ideation phase.

10. Future Scope

- Integrate with real credit bureaus (CIBIL/Experian/Equifax) for verified, bureau-grade credit scores.
- Add user authentication and role-based access so applicants can securely track their own history.
- Move server-side secrets (Claude API key, Google service account credentials) to a secure backend vault instead of the Admin panel UI.
- Extend AI Financial Tips into a fully personalized financial planning module (savings goals, investment suggestions).
- Add multi-language support to widen accessibility across diverse user bases.
- Introduce push/email notifications for EMI due dates and credit score changes.

11. Appendix

Source Code

Team GitHub repository link: *[Insert GitHub repository URL here]*

Dataset Link

Not applicable — FinWise AI uses live user-submitted data and Claude AI reasoning rather than a static training dataset. [Insert dataset link here if one is used, e.g., for score-band reference data.]

GitHub & Project Demo Link

Live Deployment: <https://sensational-selkie-59db0b.netlify.app/>

GitHub Repository: [Insert GitHub repository URL here]

Skills Utilized

- Python (Programming Language)
- CSS Animations
- JavaScript (Programming Language)
- HTML5
- Git (Version Control System)
- Generative Artificial Intelligence
- REST APIs
- JSON